

A BÉZOUT DOMAIN THAT IS NOT AN ELEMENTARY DIVISOR DOMAIN

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ABSTRACT. We settle in the negative the longstanding question whether every Bézout domain is an elementary divisor domain by constructing a Bézout domain over which an explicit 2×2 matrix has no Smith normal form. The obstruction is topological and is detected by the Möbius line bundle.

This is an early draft. The complete Lean formalization is available at <https://github.com/Zelaron/bezout-counterexample-lean>.

1. INTRODUCTION

The Smith normal form over a principal ideal domain is a fundamental structure theorem: every matrix can be diagonalized by invertible row and column operations, with diagonal entries forming a divisibility chain. In 1943, Helmer asked whether this conclusion continues to hold over every Bézout domain [4].¹ We answer this question in the negative.

Throughout, rings are commutative with identity, and ring homomorphisms preserve the identity. A *Bézout domain* is an integral domain in which every finitely generated ideal is principal; equivalently, every two-generated ideal is principal. For a ring A , two $m \times n$ matrices F and G over A are *equivalent* if $PFQ = G$ for some $P \in \mathrm{GL}_m(A)$ and $Q \in \mathrm{GL}_n(A)$. A *Smith normal form* of F is an equivalent diagonal matrix whose nonzero diagonal entries occur first and satisfy $d_1 \mid d_2 \mid \cdots \mid d_r$. A domain is an *elementary divisor domain* if every matrix over it has a Smith normal form. Every elementary divisor domain is Bézout, as follows by applying the definition to 1×2 matrices. The problem is whether the converse holds.

Helmer proved the elementary divisor theorem for adequate domains [4, Theorem 3]. Kaplansky subsequently introduced elementary divisor rings and showed that, over a domain, the problem is already detected by 2×2 matrices [6, Theorem 5.1 and the subsequent remark]. Gillman and Henriksen constructed a Bézout ring with zero divisors that is not an elementary divisor ring [3, Example 4.11], but the domain case remained open.

Our construction starts from

$$(1.1) \quad A_0 = \mathbb{Q}[x, y], \quad \Delta = x^2 + y^2 - 1, \quad M = \begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix},$$

where x and y are algebraically independent over $\mathbb{Q}$.

Theorem 1.1. *There exists a Bézout domain that is not an elementary divisor domain.*

The proof constructs an ascending sequence of smooth finitely generated factorial $\mathbb{Q}$ -domains A_n together with compact sets $K_n \subset \mathrm{Spec}(A_n)(\mathbb{R})$. Every transition map $K_{n+1} \rightarrow K_n$ is a monotone surjection, so the pullback of the Möbius line bundle on the real unit circle remains nonorientable. At the same time, every two-generated ideal occurring at some stage becomes principal at a later stage. The union $R = \bigcup_n A_n$ is therefore a Bézout domain. An equivalence of M to Smith normal form over R would already be defined at one finite stage and would yield a nowhere-zero section of the corresponding nonorientable line bundle.

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¹Helmer uses the term “Prüfer ring” for what we here call “Bézout domain”.

The algebraic point is to principalize a two-generated ideal without destroying the topological obstruction. We use the weighted invariant of Abramovich, Temkin, and Włodarczyk [1], in the presentation of Brais [2]. For a center of codimension at least two, we form its extended Rees algebra and apply Jouanolou's device. The resulting smooth factorial affine ring factors the ideal by a prime power and lowers the invariant or removes one component of its maximal locus. Divisorial centers are handled by division. On real points, the torsor step admits a compact lift with connected fibers.

Outline. Section 2 gives the topological obstruction, Section 3 records the algebraic tools, Section 4 proves the principalization extension used in the construction, and Section 5 constructs the Bézout domain and proves Theorem 1.1.

Use of AI. OpenAI's GPT-6 Astra (via Codex) was used in preparing this article and its Lean formalization. Anthropic's Claude Opus 5.5 was used to first (unsuccessfully) search for a counterexample and then to check the counterexample produced by GPT-6 Astra.

2. THE TOPOLOGICAL OBSTRUCTION

For elements $a_1, \dots, a_s$ of a ring, $(a_1, \dots, a_s)$ denotes the ideal they generate. Direct calculation from (1.1) gives

$$(2.1) \quad \det M = -\Delta, \quad (1+x, y, 1-x) = A_0, \quad M^2 - 2M = \Delta I_2.$$

The middle identity follows from $(1+x) + (1-x) = 2 \in \mathbb{Q}^\times$.

2.1. Real points. For a ring A , let $\text{Spec}(A)(\mathbb{R})$ be the set of ring homomorphisms $z : A \rightarrow \mathbb{R}$, with the coarsest topology for which every evaluation map $z \mapsto z(a)$ is continuous. A homomorphism $f : A \rightarrow B$ induces the continuous map $f^* : \text{Spec}(B)(\mathbb{R}) \rightarrow \text{Spec}(A)(\mathbb{R})$, $z \mapsto z \circ f$. If $A = \mathbb{Q}[a_1, \dots, a_m]$ is finitely generated, then $z \mapsto (z(a_1), \dots, z(a_m))$ is a homeomorphism onto the real zero set in $\mathbb{R}^m$ of the relations among the a_i . Thus $\text{Spec}(A)(\mathbb{R})$ is the usual space of real points with its Euclidean topology. In particular, it is Hausdorff, and a subset of $\text{Spec}(A)(\mathbb{R})$ is compact if and only if it is closed and the generators $a_1, \dots, a_m$ are bounded on it.

A continuous surjection $q : K' \rightarrow K$ is *monotone* if every fiber $q^{-1}(z)$ is connected. In our applications, K' and K are compact Hausdorff spaces.

2.2. The Möbius line bundle. Let K be a topological space, and let $X, Y : K \rightarrow \mathbb{R}$ be continuous functions satisfying $X^2 + Y^2 = 1$. Write $M(X, Y)$ for the matrix obtained from M by substituting X and Y for x and y . For every $z \in K$, the matrix $\frac{1}{2}M(X(z), Y(z))$ is an idempotent of rank one by (2.1). Let $L_{X,Y}(z) \subset \mathbb{R}^2$ be its image. These fibers form a real line subbundle $L_{X,Y} \subset K \times \mathbb{R}^2$. We call $L_{X,Y}$ *orientable* if it admits a continuous nowhere-zero section, that is, a continuous map $v : K \rightarrow \mathbb{R}^2$ such that $0 \neq v(z) \in L_{X,Y}(z)$ for every $z \in K$. If $q : K' \rightarrow K$ is continuous, then the pullback of $L_{X,Y}$ along q is $L_{X \circ q, Y \circ q}$.

Let

$$K_0 = \{z \in \text{Spec}(A_0)(\mathbb{R}) : z(\Delta) = 0\},$$

which is the unit circle via $z \mapsto (z(x), z(y))$, and let $L_0 = L_{x,y}$ be the corresponding line bundle. For $(x, y) = (\cos \theta, \sin \theta)$,

$$(2.2) \quad M = 2 \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \end{pmatrix}.$$

Thus L_0 is the Möbius line bundle. Indeed, a continuous nowhere-zero section, pulled back to $[0, 2\pi]$, would have the form

$$f(\theta) \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}$$

for a continuous nowhere-zero function f . Equality at the endpoints would imply $f(2\pi) = -f(0)$, contradicting the intermediate value theorem. Hence L_0 is nonorientable.

Lemma 2.1. *Let $q : K' \rightarrow K$ be a monotone surjection from a compact space to a Hausdorff space, and let $X, Y : K \rightarrow \mathbb{R}$ be continuous with $X^2 + Y^2 = 1$. If $L_{X \circ q, Y \circ q}$ is orientable, then $L_{X, Y}$ is orientable.*

Proof. Let v be a continuous nowhere-zero section of $L_{X \circ q, Y \circ q}$, and put $u = v/\|v\|$. For $z \in K$, the line $L_{X, Y}(z)$ contains exactly two unit vectors. The continuous map u therefore takes values in a two-point set on the connected fiber $q^{-1}(z)$, so it is constant there. Hence $u = \bar{u} \circ q$ for a map $\bar{u} : K \rightarrow \mathbb{R}^2$ such that $\bar{u}(z)$ is a unit vector in $L_{X, Y}(z)$. Because K' is compact and K is Hausdorff, q is a closed map and hence a quotient map. Therefore $\bar{u}$ is continuous, and it orients $L_{X, Y}$. $\square$

Equivalently, nonorientability pulls back along monotone surjections between compact Hausdorff spaces.

Lemma 2.2. *A composite of monotone surjections between compact Hausdorff spaces is monotone.*

Proof. We first note that if $q : K' \rightarrow K$ is monotone and $C \subset K$ is compact and connected, then $q^{-1}(C)$ is connected. Otherwise, write $q^{-1}(C) = F \amalg G$ with F and G nonempty and closed. Each fiber of q meets at most one of F and G , because it is connected. The compact sets $q(F)$ and $q(G)$ are therefore disjoint nonempty closed subsets whose union is C , a contradiction. Applying this observation to every fiber of a second monotone surjection proves the claim. $\square$

3. ALGEBRAIC TOOLS

A domain is *factorial* if it is a unique factorization domain (UFD). We repeatedly use finitely generated $\mathbb{Q}$ -domains that are smooth over $\mathbb{Q}$ and factorial. Such rings are Noetherian.

3.1. Factoriality. We use two standard facts. A polynomial ring over a UFD is a UFD [8, Tag 0BC1]. Nagata's criterion states that if A is a Noetherian domain and $p \in A$ is a prime element such that $A[1/p]$ is a UFD, then A is a UFD [8, Tag 0AFU]. Consequently, if A is a Noetherian UFD, then so is $A[T, T^{-1}] = A[T][1/T]$. Moreover, every prime element p of A remains prime in $A[T, T^{-1}]$, because $(A/pA)[T, T^{-1}]$ is a domain.

3.2. Jouanolou rings. Let B be a ring and let $c = (c_0, \dots, c_r) \in B^{r+1}$. Put

$$q_c = \sum_{i=0}^r c_i \sigma_i - 1 \in B[\sigma_0, \dots, \sigma_r].$$

The *Jouanolou ring* of c is

$$J_B(c) = B[\sigma_0, \dots, \sigma_r]/(q_c).$$

Let $W = \operatorname{Spec} B \setminus V(c_0, \dots, c_r)$. Over W , the row $(c_0, \dots, c_r)$ defines a surjection $\mathcal{O}_W^{\oplus(r+1)} \rightarrow \mathcal{O}_W$ with locally free kernel E_c of rank r . The morphism $\operatorname{Spec} J_B(c) \rightarrow W$ is an E_c -torsor, hence a Zariski-locally trivial affine-space bundle with affine total space. This is an explicit instance of Jouanolou's device [5, Lemme 1.5]. A real point of $J_B(c)$ is a pair (z, t) with $z \in \operatorname{Spec}(B)(\mathbb{R})$ and $t = (t_0, \dots, t_r) \in \mathbb{R}^{r+1}$ satisfying $\sum_{i=0}^r t_i z(c_i) = 1$.

Lemma 3.1. *Let B be a ring and $c \in B^{r+1}$.*

- (1) *For every homomorphism $B \rightarrow B'$, one has $J_B(c) \otimes_B B' \cong J_{B'}(c')$, where c' is the image of c . In particular, $J_B(c)/\mathfrak{a}J_B(c) \cong J_{B/\mathfrak{a}}(\bar{c})$ for every ideal $\mathfrak{a} \subset B$, and the formation of $J_B(c)$ commutes with localization of B .*
- (2) *For every $j \in \{0, \dots, r\}$, eliminating σ_j gives $J_B(c)[1/c_j] \cong B[1/c_j][\sigma_i \mid i \neq j]$.*
- (3) *If B is a domain and $c_j \neq 0$ for some j , then $J_B(c)$ is a domain and $B \rightarrow J_B(c)$ is injective.*

- (4) $J_B(c)$ is formally smooth over B . If B is a smooth finitely generated $\mathbb{Q}$ -algebra, then so is $J_B(c)$.
- (5) If B is a Noetherian UFD, c_j is a prime element of B , and $c_j \nmid c_m$ for some m , then $J_B(c)$ is a UFD.

Proof. Assertion (1) follows from the right exactness of the tensor product, and (2) from solving $q_c = 0$ for σ_j .

For (3), the constant term of q_c is -1 . For every $b \in B$, the image of q_c in $(B/bB)[\sigma_0, \dots, \sigma_r]$ is therefore a unit in the corresponding formal power series ring, and hence a nonzerodivisor in the polynomial ring. If B is a domain and $0 \neq b \in B$, it follows that multiplication by b is injective on $J_B(c)$: from $bf = gq_c$ in $B[\sigma_0, \dots, \sigma_r]$, reduction modulo b gives $g \in bB[\sigma_0, \dots, \sigma_r]$, and cancellation of b gives $f \in (q_c)$. Applying this to the powers of c_j shows that $J_B(c) \rightarrow J_B(c)[1/c_j]$ is injective. By (2), the target is a polynomial ring over the domain $B[1/c_j]$ and contains B .

For (4), put $P = B[\sigma_0, \dots, \sigma_r]$. Since q_c has constant term -1 , it is a unit in $B[[\sigma_0, \dots, \sigma_r]]$ and hence a nonzerodivisor in P . The B -derivation $\mathcal{E} = \sum_{i=0}^r \sigma_i \partial / \partial \sigma_i$ of P satisfies $\mathcal{E}(q_c) = q_c + 1$. It therefore induces a map $\Omega_{P/B} \otimes_P J_B(c) \rightarrow J_B(c)$ that sends dq_c to 1. Thus the conormal sequence

$$0 \longrightarrow (q_c)/(q_c^2) \longrightarrow \Omega_{P/B} \otimes_P J_B(c) \longrightarrow \Omega_{J_B(c)/B} \longrightarrow 0$$

is split exact, so $J_B(c)$ is formally smooth over B by the split-conormal criterion [8, Tag 031J]. If B is smooth and finitely generated over $\mathbb{Q}$, then $J_B(c)$ is formally smooth and of finite presentation over $\mathbb{Q}$, hence smooth.

For (5), the ring $J_B(c)[1/c_j]$ is a UFD by (2) and the facts recalled above. By (1), $J_B(c)/c_j J_B(c) \cong J_{B/c_j B}(\bar{c})$. This is a domain by (3), because $B/c_j B$ is a domain and $\bar{c}_m \neq 0$. By (3) again, c_j is nonzero in $J_B(c)$, so c_j is a prime element of the Noetherian domain $J_B(c)$. Nagata's criterion shows that $J_B(c)$ is a UFD. $\square$

3.3. Marked centers and the weighted invariant. Let A be a smooth finitely generated $\mathbb{Q}$ -domain and let $\mathfrak{m}$ be a maximal ideal. Then $A_{\mathfrak{m}}$ is a regular local ring of dimension $n = \dim A$, and its residue field is a finite extension of $\mathbb{Q}$. If $x = (x_1, \dots, x_n)$ is a regular system of parameters of $A_{\mathfrak{m}}$, then $dx_1, \dots, dx_n$ is a basis of $\Omega_{A_{\mathfrak{m}}/\mathbb{Q}}$, because $\mathfrak{m}/\mathfrak{m}^2 \rightarrow \Omega_{A/\mathbb{Q}} \otimes \kappa(\mathfrak{m})$ is an isomorphism in characteristic zero. Let $\partial_1, \dots, \partial_n$ be the dual derivations. We use the convention $\mathbb{N} = \{0, 1, 2, \dots\}$.

Definition 3.2. A *marked center* at $\mathfrak{m}$ is a pair $J = (x, e)$ consisting of a regular system of parameters x of $A_{\mathfrak{m}}$ and rational weights $e_1 \geq e_2 \geq \dots \geq e_n \geq 0$. For $t \in \mathbb{Q}$, its *weighted ideal of level t* is

$$F_t(J) = \left(x^\alpha : \alpha \in \mathbb{N}^n, \quad \alpha_i = 0 \text{ if } e_i = 0, \right) A_{\mathfrak{m}}.$$

$$\sum_{i=1}^n e_i \alpha_i \geq t$$

If $I \subseteq A$ is an ideal contained in $\mathfrak{m}$, then J is *admissible* for I if $IA_{\mathfrak{m}} \subseteq F_1(J)$.

In the notation of [1, 2], the marked center J is written $(x_1^{a_1}, \dots, x_k^{a_k})$, where $a_i = 1/e_i$ and k is the number of nonzero weights; the condition denoted there by $v_J(I) \geq 1$ is precisely admissibility as defined above. We regard every weight vector as an element of $\mathbb{Q}^{\mathbb{N}}$ by appending zeros. Let $\preceq$ be the opposite of the lexicographic order:

$$e \preceq e' \iff e' \leq_{\text{lex}} e.$$

Write $e \prec e'$ when $e \preceq e'$ and $e \neq e'$. This is the order obtained from the lexicographic order in [1, 2] after replacing the entries a_i by the weights $e_i = 1/a_i$; with this convention, worse singularities have larger invariants. For example, if $I = (y^2 - x^3) \subset \mathbb{Q}[x, y]$ and $\mathfrak{m} = (x, y)$, then the marked center with parameters (y, x) and weights $(\frac{1}{2}, \frac{1}{3})$ is admissible, and $\text{inv}_{\mathfrak{m}}(I) = (\frac{1}{2}, \frac{1}{3}, 0, \dots)$ corresponds to the invariant $(2, 3)$ in the notation of [1].

The formulation in [2, Sections 3–4] assumes pure positive codimension. For arbitrary nonzero ideals, we use the corresponding results in [1, Sections 5–6]. All theorem numbers for [1] refer to the published version; the numbering in arXiv:1906.07106v3 is different. For a maximal ideal $\mathfrak{m} \supseteq I$, write

$$\text{ord}_{\mathfrak{m}}(I) = \max\{b \in \mathbb{N} : IA_{\mathfrak{m}} \subseteq \mathfrak{m}^b A_{\mathfrak{m}}\}.$$

Theorem 3.3. *Let A be a smooth finitely generated $\mathbb{Q}$ -domain and let $I \subset A$ be a nonzero ideal.*

- (1) (Invariant and maximal center; [1, Theorem 5.3.1], [2, Theorem 3.1.1].) *For every maximal ideal $\mathfrak{m} \supseteq I$ there is an admissible marked center for I at $\mathfrak{m}$ whose weight vector is lexicographically smallest among all admissible marked centers at $\mathfrak{m}$, and all admissible marked centers with this weight vector have the same weighted ideals. We call this weight vector the invariant $\text{inv}_{\mathfrak{m}}(I)$, and the common weighted ideals $F_t(\mathfrak{m})$ the maximal center of I at $\mathfrak{m}$. All weights of $\text{inv}_{\mathfrak{m}}(I)$ are at most 1, and its first weight is $1/\text{ord}_{\mathfrak{m}}(I)$ [1, Section 5.1]. In particular, the first weight has the form $1/a$ for a positive integer a , and if $b \geq 1$ and $IA_{\mathfrak{m}} \subseteq \mathfrak{m}^b A_{\mathfrak{m}}$, then it is at most $1/b$.*
- (2) (Polynomial-localization invariance; a special case of [1, Theorem 5.1.3(3)] and [2, Proposition 3.5.1].) *Let $A \rightarrow B$ be a homomorphism to a smooth finitely generated $\mathbb{Q}$ -domain, let $\mathfrak{n} \supseteq IB$ be a maximal ideal of B , and put $\mathfrak{m} = \mathfrak{n} \cap A$. Suppose that, for some $r \geq 0$ and prime ideal $\mathfrak{q}$ of $A_{\mathfrak{m}}[t_1, \dots, t_r]$, there is an A -algebra isomorphism*

$$B_{\mathfrak{n}} \cong A_{\mathfrak{m}}[t_1, \dots, t_r]_{\mathfrak{q}}.$$

Then $\mathfrak{m}$ is maximal and $\text{inv}_{\mathfrak{n}}(IB) = \text{inv}_{\mathfrak{m}}(I)$. This is the special case of smooth invariance used below.

- (3) (The maximal locus; [1, Theorems 5.1.3(2) and 6.1.1], [2, Propositions 4.1.1 and 4.1.2].) *Assume $I \neq A$. Among the invariants $\text{inv}_{\mathfrak{m}}(I)$, $\mathfrak{m} \supseteq I$ maximal, there is a largest one $\text{maxinv}(I)$ for $\preceq$. There are prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ with $\mathfrak{p}_i + \mathfrak{p}_j = A$ for $i \neq j$ such that $\text{inv}_{\mathfrak{m}}(I) = \text{maxinv}(I)$ exactly when $\mathfrak{m}$ contains some $\mathfrak{p}_i$. If k is the number of nonzero entries of $\text{maxinv}(I)$ and $\mathfrak{m} \supseteq \mathfrak{p}_i$, there is a regular system of parameters x of $A_{\mathfrak{m}}$ with $\mathfrak{p}_i A_{\mathfrak{m}} = (x_1, \dots, x_k)$ such that $(x, \text{maxinv}(I))$ is admissible for I at $\mathfrak{m}$; by (1), its weighted ideals are the $F_t(\mathfrak{m})$. We call $V(\mathfrak{p}_1), \dots, V(\mathfrak{p}_r)$ the components of the maximal locus.*
- (4) (The global center; [2, Proposition 4.1.1 and Definition 4.1.4], together with affine gluing.) *Let $\mathfrak{p}$ be one of the $\mathfrak{p}_i$. For the center obtained by retaining the component $V(\mathfrak{p})$ of the global associated center, put*

$$F_t = \{f \in A : f \in F_t(\mathfrak{m}) \text{ for every maximal ideal } \mathfrak{m} \supseteq \mathfrak{p}\} \quad (t \in \mathbb{Q}).$$

Indeed, restrict Brais's global associated center to an open neighborhood of $V(\mathfrak{p})$ disjoint from the other components of the maximal locus and, level by level, glue its ideals to the unit ideal on $\text{Spec}(A) \setminus V(\mathfrak{p})$. The displayed intersection is the ideal of global sections of the resulting level- t ideal sheaf. Hence $F_t A_{\mathfrak{m}} = F_t(\mathfrak{m})$ for every maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$, and $F_t A_{\mathfrak{m}} = A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m} \not\supseteq \mathfrak{p}$.

- (5) (Well-ordering; [1, Section 5.1], [2, Proposition 3.3.3].) *There is a set Γ of weight vectors, independent of A and I , that contains all invariants and such that, for every N , its elements with at most N nonzero entries are well-ordered by $\preceq$. This is the reciprocal, zero-padded reformulation of the sets Γ_j constructed in [2, Section 3.3, especially Proposition 3.3.3].*

In (3), the maximal locus is the set of closed points of $V(\mathfrak{p}_1) \amalg \dots \amalg V(\mathfrak{p}_r)$; it is smooth of codimension k , and each $V(\mathfrak{p}_i)$ is one of its connected components.

Justification of the enlargement of Γ in (5). The set described in (5) may be larger than the set of actual invariants. We explain why it contains the latter and is still well-ordered in bounded length.

In the differential construction of [2, Sections 3.2–3.3], after weights $e_1, \dots, e_r$ have been chosen, the next positive weight has the form

$$e_{r+1} = \frac{1 - \sum_{i=1}^r \beta_i e_i}{q}, \quad \beta_i \in \mathbb{N}, \quad q \in \mathbb{N}_{>0}, \quad 0 < 1 - \sum_{i=1}^r \beta_i e_i \leq 1.$$

Let D be the product of the denominators, in lowest terms, of $e_1, \dots, e_r$. The numerator is h/D for an integer $1 \leq h \leq D$, and therefore

$$e_{r+1} = \frac{h}{Dq} = \frac{D!}{a}, \quad a = \frac{D!Dq}{h} \in \mathbb{N}_{>0}.$$

For the first weight, the empty product gives $D = 1$ and $e_1 = 1/a$.

For a fixed prefix, the possible next positive weights lie in $\{D!/a : a \in \mathbb{N}_{>0}\}$, which is well-ordered by the reverse of the usual numerical order. Allowing a zero entry, after which every entry is zero, adds a largest element for this order. In a strictly descending sequence for $\preceq$, the first entries must therefore eventually stabilize; once a prefix has stabilized, the next entries must eventually stabilize as well. For sequences of length at most N , induction through the N positions rules out an infinite strictly descending sequence. This proves the required well-ordering. $\square$

Notation 3.4. Let A be as in Theorem 3.3, let $I \subsetneq A$ be a nonzero ideal, let $V(\mathfrak{p})$ be a component of the maximal locus, and let $e = \max_{\text{inv}}(I)$ have k nonzero entries. Choose an integer $d \geq 1$ with $de_i \in \mathbb{Z}$ for all i , and index the global center of Theorem 3.3(4) by integers: $\mathcal{F}_j = \mathcal{F}_{j/d}$ for $j \in \mathbb{Z}$. Put

$$\mathcal{R} = \bigoplus_{j \in \mathbb{Z}} \mathcal{F}_j T^j \subset A[T, T^{-1}], \quad s = T^{-1},$$

$$\mathcal{R}_+ = (\mathcal{F}_j T^j : j \geq 1) \mathcal{R}, \quad I_w = (f T^d : f \in I) \mathcal{R}.$$

Thus $\mathcal{R}$ is the extended Rees algebra of the weighted center along $V(\mathfrak{p})$, s is the exceptional element, and I_w is the weak transform of I . For $1 \leq j \leq d$, let G_j be a finite set of generators of $\mathcal{F}_j$.

Lemma 3.5. *In the situation of Notation 3.4:*

- (1) $\mathcal{F}_j = A$ for $j \leq 0$, $\mathcal{F}_1 = \mathfrak{p}$, $\mathfrak{p}^j \subseteq \mathcal{F}_j$ for $j \geq 0$, and $I \subseteq \mathcal{F}_d$. In particular, $I_w \subseteq \mathcal{R}$, $I\mathcal{R} = s^d I_w$, and $\mathfrak{p} = \mathcal{F}_1 \subseteq s\mathcal{R}$.
- (2) $\mathcal{F}_j = \sum_{q=1}^d \mathcal{F}_q \mathcal{F}_{j-q}$ for $j > d$. Consequently, $\mathcal{R} = A[s, gT^j : g \in G_j, 1 \leq j \leq d]$, and the elements gT^j with $g \in G_j$ and $1 \leq j \leq d$ generate $\mathcal{R}_+$.
- (3) $\mathcal{R}[1/s] = A[T, T^{-1}]$.
- (4) Let $\mathfrak{m} \supseteq \mathfrak{p}$ be a maximal ideal, let x be as in Theorem 3.3(3), and put $w_i = de_i$ for $i \leq k$. Then $u_i \mapsto x_i T^{w_i}$ induces an isomorphism

$$A_{\mathfrak{m}}[s, u_1, \dots, u_k] / (x_i - s^{w_i} u_i \mid 1 \leq i \leq k) \cong \mathcal{R} \otimes_A A_{\mathfrak{m}}.$$

If $\mathfrak{m} \not\supseteq \mathfrak{p}$, then $\mathcal{R} \otimes_A A_{\mathfrak{m}} = A_{\mathfrak{m}}[T, T^{-1}]$. In particular, $\mathcal{R}$ is a smooth finitely generated $\mathbb{Q}$ -domain.

- (5) $\mathcal{R}/s\mathcal{R} \cong \bigoplus_{j \geq 0} \mathcal{F}_j / \mathcal{F}_{j+1}$ is a domain. Hence s is a prime element of $\mathcal{R}$.

Proof. By Theorem 3.3(4), every assertion about the ideals $\mathcal{F}_j$ may be checked after localization at maximal ideals. If $\mathfrak{m} \not\supseteq \mathfrak{p}$, then $\mathcal{F}_j A_{\mathfrak{m}} = A_{\mathfrak{m}}$ for every j . If $\mathfrak{m} \supseteq \mathfrak{p}$, then $\mathcal{F}_j A_{\mathfrak{m}} = \mathcal{F}_{j/d}(\mathfrak{m})$ is generated by the monomials x^α with $\sum_{i=1}^k w_i \alpha_i \geq j$, by Theorem 3.3(3). All weights are at most 1 by Theorem 3.3(1), so the weights w_i are integers in $\{1, \dots, d\}$. This gives $\mathcal{F}_1 A_{\mathfrak{m}} = (x_1, \dots, x_k) = \mathfrak{p} A_{\mathfrak{m}}$ and $\mathfrak{p}^j \subseteq \mathcal{F}_j$, while $I \subseteq \mathcal{F}_d$ is admissibility. Since $f = s^d \cdot f T^d$ for $f \in I$ and $g = s \cdot g T$ for $g \in \mathcal{F}_1$, we have $I\mathcal{R} = s^d I_w$ and $\mathcal{F}_1 \subseteq s\mathcal{R}$. This proves (1).

For (2), a monomial of weight at least $j > d$ is the product of some x_i , which has weight $w_i \in \{1, \dots, d\}$, and a monomial of weight at least $j - w_i$. By induction on j , every homogeneous

element of $\mathcal{R}$ of positive degree lies in the ideal generated by the elements gT^j and is a polynomial in them with coefficients in A . Moreover, $\mathcal{F}_j T^j = As^{-j}$ for $j \leq 0$. Assertion (3) holds because $\mathcal{F}_j = A$ for $j \leq 0$.

For (4), the formation of $\mathcal{R}$ commutes with localization, so $\mathcal{R} \otimes_A A_{\mathfrak{m}}$ is the extended Rees algebra of the filtration $(F_{j/d}(\mathfrak{m}))_{j \in \mathbb{Z}}$. For a filtration given by monomials in part of a regular system of parameters, the displayed presentation is [7, Proposition 5.2.2]; see also [2, Equation (2.3)]. Its left-hand side is formally smooth over $\mathbb{Q}$ by the criterion [8, Tag 031.J] already used in Lemma 3.1(4): the derivations $\partial_1, \dots, \partial_k$ of $A_{\mathfrak{m}}$, extended to $A_{\mathfrak{m}}[s, u_1, \dots, u_k]$ by $\partial_i(s) = \partial_i(u_j) = 0$, satisfy $\partial_i(x_j - s^{w_j} u_j) = \delta_{ij}$. Every localization of $\mathcal{R}$ at a prime ideal is a localization of some $\mathcal{R} \otimes_A A_{\mathfrak{m}}$, so it is formally smooth over $\mathbb{Q}$. Since $\mathcal{R}$ is of finite type over $\mathbb{Q}$ by (2), it is smooth. It is a domain because it is contained in $A[T, T^{-1}]$.

For (5), the identification $\mathcal{R}/s\mathcal{R} \cong \bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ follows from $s\mathcal{R} \cap \mathcal{F}_j T^j = \mathcal{F}_{j+1} T^j$. By (4), for every maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$ the graded ring $\bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ becomes the polynomial ring $(A/\mathfrak{p})_{\mathfrak{m}/\mathfrak{p}}[u_1, \dots, u_k]$ after localization at $\mathfrak{m}$. In particular, each $\mathcal{F}_j/\mathcal{F}_{j+1}$ is a locally free $A/\mathfrak{p}$ -module, hence torsion-free. Now let $f \in \mathcal{F}_a \setminus \mathcal{F}_{a+1}$ and $g \in \mathcal{F}_b \setminus \mathcal{F}_{b+1}$. Their images in $(\mathcal{F}_a/\mathcal{F}_{a+1})_{\mathfrak{m}}$ and $(\mathcal{F}_b/\mathcal{F}_{b+1})_{\mathfrak{m}}$ are nonzero at every maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$. Their product in $(A/\mathfrak{p})_{\mathfrak{m}/\mathfrak{p}}[u_1, \dots, u_k]$ is therefore nonzero, so $fg \notin \mathcal{F}_{a+b+1}$. Thus $\bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ is a domain, and it is nonzero because $\mathcal{F}_1 = \mathfrak{p} \neq A$. $\square$

Theorem 3.6 (Strict decrease of the invariant). *In the situation of Notation 3.4, let P be a maximal ideal of $\mathcal{R}$ that contains $I_w + s\mathcal{R}$ but not $\mathcal{R}_+$. Then $\text{inv}_P(I_w) \prec \text{maxinv}(I)$.*

Proof. We use the affine extended-Rees argument underlying property (C) in [2, Section 2.2]. Let $\mathfrak{m} = P \cap A$. The weak Nullstellensatz makes $\mathfrak{m}$ maximal, and $\mathfrak{p} \subseteq \mathfrak{m}$ because $\mathfrak{p}\mathcal{R} \subseteq s\mathcal{R} \subseteq P$. Work near $\mathfrak{m}$ with compatible parameters as in Lemma 3.5(4). Let P_0 be the vertex point over $\mathfrak{m}$, defined by $s = u_1 = \dots = u_k = 0$ and the remaining parameters of $A_{\mathfrak{m}}$. Equivalently,

$$P_0 = \left\{ \sum_j f_j T^j \in \mathcal{R} : f_0 \in \mathfrak{m} \right\}.$$

The differential construction of the invariant at the vertex gives $\text{inv}_{P_0}(I_w) = e := \text{maxinv}(I)$: the positive weights are those of the original center, now on the u_i , with s of weight zero. This follows by transporting each step of the differential construction through $x_i = s^{w_i} u_i$; admissibility and the successive least derivative candidates are preserved. In this maximal center every homogeneous element fT^j , $f \in \mathcal{F}_j$, $j \geq 1$, belongs to the weighted ideal of level $1/d$.

Choose such an element $h = fT^j \notin P$. Spreading the vertex center and this membership to a principal neighborhood $D(G)$ of P_0 , the local construction of the maximal center gives the following: at every maximal ideal $Q \in D(G)$ containing I_w , one has $\text{inv}_Q(I_w) \leq e$, and equality implies $h \in Q$. Indeed, in the equality case the spread center is the maximal center at Q by uniqueness, and its positive-level weighted ideals lie in Q .

The grading acts by automorphisms $\sigma_\lambda(T) = \lambda T$, $\lambda \in \mathbb{Q}^\times$, and preserves I_w . Since $s \in P$, all negative-degree terms of G vanish modulo P . The constant term of $\sigma_\lambda(G)$ modulo P is nonzero, because $G \notin P_0$. Thus the image of $\sigma_\lambda(G)$ is a nonzero polynomial in λ over the field $\mathcal{R}/P$, and some $\lambda \in \mathbb{Q}^\times$ makes it nonzero. Then $Q = \sigma_\lambda^{-1}(P)$ lies in $D(G)$, contains I_w , and has the same invariant as P . But $\sigma_\lambda(h) = \lambda^j h \notin P$, so $h \notin Q$. Equality with e is impossible, and $\text{inv}_P(I_w) \prec e$ follows. $\square$

Lemma 3.7. *Let $\mathfrak{m} \supseteq I$ be a maximal ideal of A , and let δ be a $\mathbb{Q}$ -derivation of A with $\delta(I) \subseteq I$.*

- (1) *The derivation δ preserves the maximal center: $\delta(F_t(\mathfrak{m})) \subseteq F_t(\mathfrak{m})$ for every t . Consequently, $\delta(\mathcal{F}_j) \subseteq \mathcal{F}_j$ in Notation 3.4.*

- (2) Suppose that $\delta_1, \dots, \delta_m$ are derivations of A with $\delta_j(I) \subseteq I$, and that $\det(\delta_j(z_q))_{1 \leq j, q \leq m} \notin \mathfrak{m}$ for some $z_1, \dots, z_m \in A$. Then $\text{inv}_{\mathfrak{m}}(I)$ has at most $\dim A - m$ nonzero entries.

Proof. For (1), put $C = A_{\mathfrak{m}}$, write $e = \text{inv}_{\mathfrak{m}}(I)$, and let k be the number of its nonzero entries. Choose parameters $x = (x_1, \dots, x_n)$ compatible with the maximal center and an integer $d \geq 1$ such that every de_i is integral. Choose an integer $D > d$. Put

$$B = A[\tau], \quad \mathfrak{n} = \mathfrak{m}B + \tau B, \quad I_D = IB + \tau^D B.$$

Let $G_t \subseteq B_{\mathfrak{n}}$ be the weighted ideal for the ordered parameters

$$(x_1, \dots, x_k, \tau, x_{k+1}, \dots, x_n)$$

with weights $(e_1, \dots, e_k, 1/D, 0, \dots, 0)$.

Claim. The filtration (G_t) is the maximal center of I_D at $\mathfrak{n}$.

Proof of the claim. Write $a_i = 1/e_i$, so $a_i \leq d < D$, and use the differential construction of [2, Sections 3.2–3.3]. After a prefix $a_1, \dots, a_r$ has been chosen, the next exponent is the least candidate of the form

$$\frac{\sum_{j>r} \beta_j}{1 - \sum_{i=1}^r \beta_i/a_i}, \quad 1 - \sum_{i=1}^r \beta_i/a_i > 0,$$

for which the corresponding derivative ideal is the unit ideal; the variables include τ . For $IB_{\mathfrak{n}}$, Theorem 3.3(1)–(2) shows that the maximal center is the smooth pullback of the original center, with τ of weight zero, so its positive exponents are $a_1, \dots, a_k$. At every stage, the derivative ideal for $I_D B_{\mathfrak{n}} = IB_{\mathfrak{n}} + \tau^D B_{\mathfrak{n}}$ is the sum of the derivative ideals for the two summands. The derivative ideal contributed by $\tau^D B_{\mathfrak{n}}$ is proper unless $\beta_{\tau} \geq D$, and in that case the displayed candidate is at least D . Since the sum of two proper ideals in the local ring $B_{\mathfrak{n}}$ is proper, adjoining τ^D does not alter the first k exponents, all of which are at most $d < D$. After those steps, $IB_{\mathfrak{n}}$ is admissible for the pulled-back center, while τ^D contributes the next exponent D and the parameter τ , obtained by differentiating $D - 1$ times. The enlarged center is admissible for $I_D B_{\mathfrak{n}}$, so the construction stops. Uniqueness of the maximal center identifies its filtration with (G_t) , proving the claim.

Extend δ to C and set $\delta(\tau) = 0$. Under the identification

$$B_{\mathfrak{n}}/\tau^D B_{\mathfrak{n}} \cong C[\tau]/(\tau^D),$$

the formulas

$$\Phi(\tau) = \tau, \quad \Phi(f) = \sum_{j=0}^{D-1} \frac{\tau^j \delta^j(f)}{j!} \quad (f \in C)$$

define an automorphism; its inverse is obtained by replacing τ by $-\tau$. Since $\delta(IC) \subseteq IC$, the automorphism Φ preserves the image of I_D . Put

$$x'_i = \sum_{j=0}^{D-1} \frac{\tau^j \delta^j(x_i)}{j!}.$$

Modulo $(\mathfrak{n}B_{\mathfrak{n}})^2$, each x'_i differs from x_i by a scalar multiple of τ . Hence

$$(x'_1, \dots, x'_k, \tau, x'_{k+1}, \dots, x'_n)$$

is a regular system of parameters. Let G'_t be the weighted ideals defined by these parameters and the same weights as G_t . Modulo τ^D , the automorphism Φ preserves the image of I_D and carries the image of G_1 to that of G'_1 . Therefore

$$I_D B_{\mathfrak{n}} \subseteq G'_1 + \tau^D B_{\mathfrak{n}} = G'_1,$$

where the equality follows from $\tau^D \in G'_1$. Thus the new center is admissible for I_D . The claim identifies its weight vector as maximal, so uniqueness gives $G'_t = G_t$ for every t .

In particular, $x'_i \in G_{e_i}$ for $1 \leq i \leq k$. Under the natural map $B_{\mathfrak{n}} \rightarrow C[[\tau]]$, expand an element of G_{e_i} in the monomial generators of that ideal. Only generators of τ -degree zero or one can contribute to the coefficient of τ . In the first case the x -weight is at least e_i , and in the second it is at least $e_i - 1/D$. Hence the coefficient belongs to $F_{e_i-1/D}(\mathfrak{m})$, and therefore

$$\delta(x_i) \in F_{e_i-1/D}(\mathfrak{m}) = F_{e_i}(\mathfrak{m}).$$

The equality holds because e_i and every x -monomial weight lie in $\frac{1}{d}\mathbb{Z}$, whereas $1/D < 1/d$. The Leibniz rule now gives $\delta(F_t(\mathfrak{m})) \subseteq F_t(\mathfrak{m})$ for every t . Applying this at every maximal ideal containing the chosen component and using the definition in Theorem 3.3(4) proves $\delta(\mathcal{F}_j) \subseteq \mathcal{F}_j$.

For (2), let k be the number of nonzero entries of $e = \text{inv}_{\mathfrak{m}}(I)$. For $0 < \varepsilon \leq e_k$ we have $F_{\varepsilon}(\mathfrak{m}) = (x_1, \dots, x_k)$, so by (1) every $\delta_j(x_i)$ with $i \leq k$ lies in $\mathfrak{m}$. Writing $\delta_j = \sum_i \delta_j(x_i) \partial_i$, we see that the matrix $(\delta_j(z_l))$ is the product of $(\delta_j(x_i))_{j,i}$ and $(\partial_i z_l)_{i,l}$. Modulo $\mathfrak{m}$, the first factor has zero columns for $i \leq k$, so the product has rank at most $n - k$ over $\kappa(\mathfrak{m})$. Hence $m \leq n - k$. $\square$

4. PRINCIPALIZING IDEALS WHILE PRESERVING THE OBSTRUCTION

Throughout this section, A is a smooth finitely generated factorial $\mathbb{Q}$ -domain, and $I \subsetneq A$ is a nonzero ideal. We fix a component $V(\mathfrak{p})$ of the maximal locus of I and use Notation 3.4; in particular, $k \geq 1$ is the number of nonzero entries of $e = \text{maxinv}(I)$. Since A is a UFD and $\mathfrak{p} \neq 0$, the prime ideal $\mathfrak{p}$ contains a prime element π . Let $c(I)$ denote the number of components of the maximal locus.

One step replaces (A, I) by a pair (B, I_1) , where $B = A$ or B is a smooth finitely generated factorial $\mathbb{Q}$ -domain containing A , and factors IB as $IB = \alpha I_1$ for some $\alpha \in B$. The complexity decreases in the following sense: either $I_1 = B$, or $\text{maxinv}(I_1) \prec \text{maxinv}(I)$, or $\text{maxinv}(I_1) = \text{maxinv}(I)$ and $c(I_1) < c(I)$. We distinguish the divisorial case $k = 1$ from the case $k \geq 2$.

4.1. The divisorial step.

Lemma 4.1. *Assume $k = 1$, so $e = (1/a, 0, 0, \dots)$ for a positive integer a . Then $\mathfrak{p} = \pi A$, $I \subseteq \pi^a A$, and for $I_1 = (I : \pi^a)$ one has $I = \pi^a I_1$. Moreover, either $I_1 = A$, or $\text{maxinv}(I_1) \prec \text{maxinv}(I)$, or $\text{maxinv}(I_1) = \text{maxinv}(I)$ and $c(I_1) < c(I)$.*

Proof. At every maximal ideal containing $\mathfrak{p}$, the ideal $\mathfrak{p}$ is generated by one parameter. Thus $\mathfrak{p}$ is a height-one prime, hence $\mathfrak{p} = \pi A$ because A is factorial. Admissibility gives $I \subseteq \pi^a A$, and therefore $I = \pi^a I_1$.

No maximal ideal can contain both I_1 and π : otherwise $IA_{\mathfrak{m}} \subseteq \mathfrak{m}^{a+1}A_{\mathfrak{m}}$, so the first weight of $\text{inv}_{\mathfrak{m}}(I)$ would be at most $1/(a+1)$, contradicting $\text{inv}_{\mathfrak{m}}(I) = e$. Hence π is a unit at every maximal ideal containing I_1 , where I_1 and I have the same localization and the same invariant. Thus the chosen component $V(\mathfrak{p})$ disappears and no new maximal component is created. The conclusion follows from Theorem 3.3(3). $\square$

4.2. The torsor step. Assume $k \geq 2$. Let

$$\mathbf{h} = (h_0, h_1, \dots, h_{\ell}) = (\pi T, gT^j : g \in G_j, 1 \leq j \leq d)$$

be the tuple of elements of $\mathcal{R}_+$ obtained by listing πT followed by the elements gT^j with $g \in G_j$ and $1 \leq j \leq d$. Write $h_i = g_i T^{j_i}$, where $1 \leq j_i \leq d$ and $g_i \in \mathcal{F}_{j_i}$; thus $g_0 = \pi$ and $j_0 = 1$. By Lemma 3.5(2), the entries of $\mathbf{h}$ generate $\mathcal{R}_+$. Put

$$U = J_{\mathcal{R}}(\mathbf{h}) = \mathcal{R}[\sigma_0, \dots, \sigma_{\ell}] / \left(\sum_{i=0}^{\ell} h_i \sigma_i - 1 \right),$$

so that $\text{Spec } U \rightarrow \text{Spec } \mathcal{R} \setminus V(\mathcal{R}_+)$ is the Jouanolou affine-space bundle associated with $\mathbf{h}$, and let $I_1 = I_w U$.

Lemma 4.2. *The ring U is a smooth finitely generated factorial $\mathbb{Q}$ -domain, the homomorphism $A \rightarrow U$ is injective, and $IU = s^d I_1$.*

Proof. By Lemmas 3.5 and 3.1, the ring U is a smooth finitely generated $\mathbb{Q}$ -domain, $\mathcal{R} \rightarrow U$ is injective, and $IU = s^d I_1$. Modulo s one has

$$U/sU \cong J_{\mathcal{R}/s\mathcal{R}}(\bar{\mathbf{h}}).$$

The base ring is a domain by Lemma 3.5(5), and the row $\bar{\mathbf{h}}$ is nonzero: after localizing at a maximal ideal over $\mathfrak{p}$ and writing $w_1 = de_1$, one has $x_1 \in \mathcal{F}_{w_1} A_{\mathfrak{m}} \setminus \mathcal{F}_{w_1+1} A_{\mathfrak{m}}$, so some generator in G_{w_1} has nonzero initial form. Lemma 3.1(3) therefore shows that U/sU is a domain, so s is prime in U .

After inverting s ,

$$U[1/s] \cong J_{A[T, T^{-1}]}(\mathbf{h}).$$

The Laurent polynomial ring is factorial, the row contains the prime element πT , and it also contains an element gT not divisible by πT : otherwise the height- k prime $\mathfrak{p}$, with $k \geq 2$, would be generated by π . Hence Lemma 3.1(5) makes $U[1/s]$ factorial, and Nagata's criterion makes U factorial. The inclusion $A \hookrightarrow U$ follows from $A \subseteq \mathcal{R} \hookrightarrow U$. $\square$

Lemma 4.3. *The ideal I_1 is nonzero. Moreover, either $I_1 = U$, or $\text{maxinv}(I_1) \prec \text{maxinv}(I)$, or $\text{maxinv}(I_1) = \text{maxinv}(I)$ and $c(I_1) < c(I)$.*

Proof. The ideal I_1 contains $fT^d \neq 0$ for $0 \neq f \in I$, since $\mathcal{R} \rightarrow U$ is injective. Let $Q \supseteq I_1$ be maximal and put $P = Q \cap \mathcal{R}$. The contractions P and $Q \cap A$ are maximal by the weak Nullstellensatz. Since the h_i generate the unit ideal in U , some $h_i \notin P$; hence U_Q is a localization of a polynomial ring over $\mathcal{R}_P$, and

$$\text{inv}_Q(I_1) = \text{inv}_P(I_w).$$

If $s \in Q$, Theorem 3.6 gives a strict decrease. If $s \notin Q$, then $\mathcal{R}_P$ is a localization of $A_{Q \cap A}[T, T^{-1}]$ by Lemma 3.5(3), so Theorem 3.3(2) gives

$$\text{inv}_Q(I_1) = \text{inv}_{Q \cap A}(I).$$

Moreover $Q \cap A$ cannot contain $\mathfrak{p}$, for then $s^{j_i} h_i = g_i \in \mathfrak{p} \subseteq P$ would imply $h_i \in P$ for every i .

It follows that every invariant of I_1 is at most $\text{maxinv}(I)$, and equality can occur only over a component $V(\mathfrak{p}')$ of the old maximal locus with $\mathfrak{p}' \neq \mathfrak{p}$. Conversely, let Q be any maximal ideal containing $\mathfrak{p}'U$. Then $s \notin Q$: otherwise $\mathfrak{p}U \subseteq sU \subseteq Q$, contradicting $(\mathfrak{p} + \mathfrak{p}')U = U$. Since $I \subseteq \mathfrak{p}'$, the identity $IU = s^d I_1$ gives $I_1 U_Q = IU_Q \subseteq QU_Q$, so $I_1 \subseteq Q$ and the invariant at Q equals $\text{maxinv}(I)$. Finally,

$$\mathcal{R}/\mathfrak{p}'\mathcal{R} \cong (A/\mathfrak{p}')[T, T^{-1}],$$

and some generator from G_1 survives modulo $\mathfrak{p}'$, so Lemma 3.1(3) shows that $U/\mathfrak{p}'U$ is a nonzero domain. Thus the ideals $\mathfrak{p}'U$, for the old components other than $\mathfrak{p}$, are proper pairwise-comaximal primes and are exactly the components that can remain. The claimed decrease follows. $\square$

4.3. Termination. The torsor step raises dimension, so we keep track of the new smooth directions. Fix $N \geq 0$. Say that a triple $(B, \mathfrak{b}, m)$ satisfies $(\star_N)$ if

$$\dim B \leq N + m$$

and, at every maximal ideal $\mathfrak{m} \supseteq \mathfrak{b}$, there are m derivations preserving $\mathfrak{b}$ whose values on m elements have determinant outside $\mathfrak{m}$. By Lemma 3.7(2), every invariant of such an ideal has at most N nonzero entries. The initial triple $(A, I, 0)$ satisfies $(\star_{\dim A})$.

Lemma 4.4. *Suppose that (A, I, m) satisfies $(\star_N)$. In the divisorial step, (A, I_1, m) satisfies $(\star_N)$. In the torsor step, $(U, I_1, m + \ell + 1)$ satisfies $(\star_N)$.*

Proof. In the divisorial step, the dimension bound is unchanged and $V(I_1) \subseteq V(I)$. By Lemma 3.7(1), a derivation preserving I satisfies $\delta(\pi) = b\pi$ and therefore preserves $(I : \pi^a)$, since

$$\pi^a \delta(f) = \delta(\pi^a f) - ab \pi^a f \in I \quad (\pi^a f \in I).$$

In the torsor step, $\dim U = \dim A + 1 + \ell \leq N + m + 1 + \ell$. Let $Q \supseteq I_1$ be maximal. Then $Q \cap A$ is maximal and contains I , so choose the m derivations and elements supplied by $(\star_N)$ there, and choose i_0 with $h_{i_0} \notin Q$. Lemma 3.7(1) lets each old derivation act coefficientwise on $\mathcal{R}$ and extend to U by

$$\delta(\sigma_i) = -\sigma_i \sum_r \sigma_r \delta(h_r).$$

Together with the Euler derivation

$$E(fT^q) = qfT^q, \quad E(\sigma_i) = -j_i \sigma_i$$

and the ℓ vertical derivations

$$h_r \frac{\partial}{\partial \sigma_{i_0}} - h_{i_0} \frac{\partial}{\partial \sigma_r} \quad (r \neq i_0),$$

these give $m + \ell + 1$ derivations preserving I_1 . On the elements $z_1, \dots, z_m, h_{i_0}$ and σ_r for $r \neq i_0$, their value matrix is block upper triangular with determinant

$$\det(\delta_j(z_q)) j_{i_0} h_{i_0} (-h_{i_0})^\ell \notin Q.$$

Thus $(U, I_1, m + \ell + 1)$ satisfies $(\star_N)$. □

4.4. Real points.

Lemma 4.5. *In the torsor step, let $K \subset \text{Spec}(A)(\mathbb{R})$ be compact, put $E_i = 2d!/j_i$ (an even integer), and let $K' \subset \text{Spec}(U)(\mathbb{R})$ be the set of real points z' such that*

$$\begin{aligned} z'|_A \in K, \quad z'(s) &\geq 0, \quad \sum_{i=0}^{\ell} z'(h_i)^{E_i} = 1, \\ z'(\sigma_i) &= z'(h_i)^{E_i-1} \quad (0 \leq i \leq \ell). \end{aligned}$$

Then K' is compact, and $K' \rightarrow K$ is a monotone surjection.

Proof. The last two conditions are compatible with $\sum_i h_i \sigma_i = 1$. If $z' \in K'$ and $z = z'|_A$, then $s^{j_i} h_i = g_i$ and $j_i E_i = 2d!$ give

$$(4.1) \quad z'(s)^{2d!} = \sum_{i=0}^{\ell} z(g_i)^{E_i} =: \nu(z).$$

The defining conditions are closed, $|z'(h_i)|, |z'(\sigma_i)| \leq 1$, and (4.1) bounds $z'(s)$ on K' . Since U is generated over A by s , the h_i , and the σ_i , while elements of A are bounded on the compact set K , the criterion of Subsection 2.1 shows that K' is compact. The restriction map $K' \rightarrow K$ is continuous.

The function $\nu(z)$ is positive exactly when $\mathfrak{p} \not\subseteq \ker z$: every g_i lies in $\mathfrak{p}$, while the coefficients with $j_i = 1$ generate $\mathfrak{p}$. If $\nu(z) > 0$, equation (4.1) forces $z'(s) = \rho := \nu(z)^{1/(2d!)} > 0$. There is then a unique point over z : extend z to $A[T, T^{-1}]$ by $z'(T) = \rho^{-1}$ and put $z'(\sigma_i) = z'(h_i)^{E_i-1}$.

Now suppose $\mathfrak{p} \subseteq \ker z$. Then every point over z sends s to zero. Choose a maximal ideal $\mathfrak{m} \supseteq \ker z$ and use the local presentation in Lemma 3.5(4). Points of $\mathcal{R}$ over z with $s = 0$ are parametrized by $a = (a_1, \dots, a_k) \in \mathbb{R}^k$, where a_r is the value of $x_r T^{w_r}$. Write $H_i(a)$ for the value of h_i at this point. Each H_i is weighted homogeneous of degree j_i , so, with

$$\lambda \star a = (\lambda^{w_1} a_1, \dots, \lambda^{w_k} a_k), \quad \varphi(a) = \sum_i H_i(a)^{E_i},$$

one has $\varphi(\lambda \star a) = \lambda^{2d} \varphi(a)$. Moreover, if $\varphi(a) = 0$, then the even exponents force every $H_i(a)$ to vanish. The corresponding point therefore vanishes on the ideal $\mathcal{R}_+$ generated by the h_i , including every $x_r T^{w_r}$, and hence $a = 0$.

The fiber of $K' \rightarrow K$ over z is therefore the level set $\varphi(a) = 1$, with the values of the σ_i determined by the defining equations. The normalization

$$a \longmapsto \varphi(a)^{-1/(2d)} \star a$$

is a continuous surjection from $\mathbb{R}^k \setminus \{0\}$ onto this fiber. Since $k \geq 2$, the source is connected. Thus every fiber is nonempty and connected, and $K' \rightarrow K$ is a monotone surjection. $\square$

4.5. The principalization result.

Proposition 4.6 (Principalization extension). *Let A be a smooth finitely generated factorial $\mathbb{Q}$ -domain, let $a, b \in A$ be not both zero, and let $K \subset \text{Spec}(A)(\mathbb{R})$ be compact. There are a smooth finitely generated factorial $\mathbb{Q}$ -domain A' , an injection $A \hookrightarrow A'$, and a compact set $K' \subset \text{Spec}(A')(\mathbb{R})$ such that $(a, b)A'$ is principal and the induced map $K' \rightarrow K$ is a monotone surjection.*

Proof. Put $I = (a, b)$ and $N = \dim A$. We prove the following auxiliary assertion. Let B be a smooth finitely generated factorial $\mathbb{Q}$ -domain, let $\mathfrak{b} \subseteq B$ be a nonzero ideal, let $m \geq 0$ be such that $(B, \mathfrak{b}, m)$ satisfies $(\star_N)$, and let $K \subseteq \text{Spec}(B)(\mathbb{R})$ be compact. Then there are an injection $B \hookrightarrow B'$ into a smooth finitely generated factorial $\mathbb{Q}$ -domain and a compact set $K' \subseteq \text{Spec}(B')(\mathbb{R})$ such that $\mathfrak{b}B'$ is principal and $K' \rightarrow K$ is a monotone surjection. The proposition is the initial case $(B, \mathfrak{b}, m) = (A, I, 0)$.

If $\mathfrak{b} = B$, take $B' = B$ and $K' = K$. For $\mathfrak{b} \neq B$ we argue by well-founded induction on the pair $(\max_{\text{inv}}(\mathfrak{b}), c(\mathfrak{b}))$, ordered lexicographically, with $\preceq$ on the first entry and the usual order on $\mathbb{N}$ on the second. This is well-founded by Theorem 3.3(5), because all invariants of ideals in triples satisfying $(\star_N)$ have at most N nonzero entries. Let k be the number of nonzero entries of $\max_{\text{inv}}(\mathfrak{b})$, choose a component of the maximal locus of $\mathfrak{b}$, and apply the divisorial step if $k = 1$ or the torsor step if $k \geq 2$. This gives a smooth finitely generated factorial $\mathbb{Q}$ -domain B_1 with an injection $B \hookrightarrow B_1$ (namely $B_1 = B$, or $B_1 = U$ by Lemma 4.2), a compact set $K_1 \subset \text{Spec}(B_1)(\mathbb{R})$ with $K_1 \rightarrow K$ a monotone surjection (with $K_1 = K$ in the divisorial case and K_1 as in Lemma 4.5 in the torsor case), an ideal $\mathfrak{b}_1 \subseteq B_1$, and an element $\alpha \in B_1$ with $\mathfrak{b}B_1 = \alpha \mathfrak{b}_1$ (where $\alpha = \pi^a$ or $\alpha = s^d$). Let $m_1 = m$ in the divisorial step and $m_1 = m + \ell + 1$ in the torsor step. By Lemmas 4.1, 4.3 and 4.4, the triple $(B_1, \mathfrak{b}_1, m_1)$ satisfies $(\star_N)$, and either $\mathfrak{b}_1 = B_1$ or $(\max_{\text{inv}}(\mathfrak{b}_1), c(\mathfrak{b}_1))$ is strictly smaller than $(\max_{\text{inv}}(\mathfrak{b}), c(\mathfrak{b}))$ in this lexicographic order.

In either case, the assertion for $(B_1, \mathfrak{b}_1, m_1)$ and K_1 holds, trivially if $\mathfrak{b}_1 = B_1$ and by induction otherwise. It gives $B_1 \hookrightarrow B'$ and a compact set $K' \subset \text{Spec}(B')(\mathbb{R})$ with $\mathfrak{b}_1 B' = \alpha' B'$ and $K' \rightarrow K_1$ a monotone surjection. Then $\mathfrak{b}B' = \alpha \alpha' B'$ is principal, $B \rightarrow B'$ is injective, and $K' \rightarrow K$ is a monotone surjection by Lemma 2.2. $\square$

5. CONSTRUCTION OF THE BÉZOUT DOMAIN

Construction 5.1. Fix a bijection $\langle \cdot, \cdot \rangle : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ with $i \leq \langle i, j \rangle$ for all i, j , for instance the Cantor pairing. Starting with $A_0 = \mathbb{Q}[x, y]$ and the circle K_0 , we construct injections

$$A_0 \hookrightarrow A_1 \hookrightarrow A_2 \hookrightarrow \cdots$$

of smooth finitely generated factorial $\mathbb{Q}$ -domains and compact sets $K_n \subset \text{Spec}(A_n)(\mathbb{R})$ such that each induced map $K_{n+1} \rightarrow K_n$ is a monotone surjection.

Every finitely generated $\mathbb{Q}$ -algebra is countable. Once A_i has been constructed, choose a surjection

$$\eta_i : \mathbb{N} \longrightarrow A_i \times A_i.$$

To pass from A_n to A_{n+1} , write $n = \langle i, j \rangle$. Then $i \leq n$, and we let (a, b) be the image of the pair $\eta_i(j)$ in A_n . If $a = 0$ or $b = 0$, the ideal $(a, b)A_n$ is principal, and we put $A_{n+1} = A_n$ and $K_{n+1} = K_n$. Otherwise, write $a = ca'$ and $b = cb'$ with a' and b' coprime, which is possible because A_n is a UFD. Apply Proposition 4.6 to a' , b' , and the compact set K_n , and let A_{n+1} and K_{n+1} be the resulting ring and compact set. Then $(a, b)A_{n+1} = c(a', b')A_{n+1}$ is principal.

Identify every A_n with its image in later rings and set

$$(5.1) \quad R = \bigcup_{n \geq 0} A_n.$$

Proposition 5.2. *The ring R is a Bézout domain.*

Proof. Since all transition maps are injective, the increasing union R is a domain. Let $a, b \in R$. Choose i such that $a, b \in A_i$, and choose $j \in \mathbb{N}$ with $\eta_i(j) = (a, b)$. This pair is processed while passing from A_n to A_{n+1} , where $n = \langle i, j \rangle$. Hence

$$(a, b)A_{n+1} = gA_{n+1}$$

for some $g \in A_{n+1}$. Extending to R gives $(a, b)R = gR$. Thus R is a Bézout domain. $\square$

The functions x and y on K_n are pulled back from K_0 , so the line bundle

$$L_n = L_{x,y} \quad \text{on } K_n$$

is the pullback of L_0 along the composite of the transition maps. By Lemmas 2.1 and 2.2,

$$(5.2) \quad L_n \text{ is nonorientable for every } n \geq 0.$$

Lemma 5.3. *There are no elements $p, q, u, v, \mu \in R$ such that*

$$\mu(p, q) M \begin{pmatrix} u \\ v \end{pmatrix} = 1.$$

Proof. Suppose such elements exist, and choose N such that all five lie in A_N . The identity already holds in A_N , because $A_N \hookrightarrow R$ is injective. For $z \in K_N$, put

$$\xi(z) = M(z) \begin{pmatrix} u(z) \\ v(z) \end{pmatrix} \in L_N(z).$$

The assumed identity gives $\mu(z)(p(z), q(z)) \xi(z) = 1$, so $\xi(z) \neq 0$. Thus ξ is a continuous nowhere-zero section of L_N , contradicting (5.2). $\square$

Proof of Theorem 1.1. Take the ring R from Construction 5.1. By Proposition 5.2, it is a Bézout domain. It remains to show that it is not an elementary divisor domain.

Suppose, to the contrary, that the matrix M has a Smith normal form over R :

$$(5.3) \quad PMQ = \text{diag}(d_1, d_2), \quad P, Q \in \text{GL}_2(R), \quad d_1 \mid d_2.$$

Multiplication by invertible matrices preserves the ideal generated by all matrix entries. By (2.1), the entries of M generate R . Hence $(d_1, d_2) = R$. Since $d_1 \mid d_2$, this ideal equals (d_1) , so d_1 is a unit.

Let (p, q) be the first row of P and $(u, v)^\top$ the first column of Q . Comparing upper-left entries in (5.3) gives

$$(p, q)M(u, v)^\top = d_1.$$

Multiplying by d_1^{-1} contradicts Lemma 5.3. Thus M has no Smith normal form over R , so R is not an elementary divisor domain. $\square$

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